

**A Capstone Project Report on  
Surface Water and Drinking Water Contaminants**

**CS 896: Data Science Capstone**

**By  
Kavya Sri Modepu  
Bindu Nagasai**

**Under the guidance of  
Ast. Prof. Dr. Lokesh Das**

**Wichita State University**

## Abstract

Water quality is an important environmental and public health concern, especially in regions where communities depend on both surface water and groundwater for drinking purposes. This project focuses on analyzing contaminant patterns in drinking water systems across Kansas and examining how these patterns relate to surface water conditions.

Two datasets were used in this study: a municipal drinking water dataset and a surface water dataset obtained from the Water Quality Portal (WQP). The drinking water data required significant preprocessing, including unit standardization, handling of non-numeric values, and creation of derived variables such as standardized concentration (ppb), exceedance indicators, and outlier flags.

Exploratory analysis identified Barium, Fluoride, Selenium, Nitrate, and Chromium as the most frequently reported contaminants. Among these, nitrate showed significantly higher concentration levels compared to others, increasing from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. Regression analysis indicated that year alone does not explain variation when all contaminants are combined, highlighting the importance of contaminant-specific analysis.

The surface water dataset enabled spatial analysis due to the availability of geographic coordinates. Monitoring locations were mapped, and nitrate was analyzed across locations and years. The results showed that nitrate concentrations vary across sites and fluctuate over time without a clear linear trend. Site-level analysis further confirmed that higher values are distributed across multiple locations rather than concentrated in a single area.

A key limitation of this study is the absence of geographic identifiers in the drinking water dataset, which prevents direct linkage between surface water sources and drinking water systems. Therefore, the findings are presented as two related analyses rather than a direct causal relationship.

Overall, this project helped us understand how water quality data behaves in real-world scenarios. It also showed that data limitations can affect how analysis is performed and interpreted. The results provide a useful starting point for further work on understanding water quality patterns.

## Introduction

In this project, we analyzed water quality data in Kansas to understand how contaminants behave in drinking water and surface water systems. The main idea was to see whether contamination in surface water could be related to what is observed in drinking water.

At the beginning, we expected to directly connect both datasets using location information. However, during the analysis, we noticed that the drinking water dataset does not include latitude or longitude. Because of this, it was not possible to directly link the two datasets.

Due to this limitation, we decided to approach the project in two parts. First, we analyzed contaminant patterns in the drinking water dataset. Then, we performed spatial and temporal analysis on the surface water dataset separately.

Even though a direct connection was not possible, both analyses helped us understand important patterns in the data, especially for nitrate, which appeared to behave differently compared to other contaminants.

## **Problem Statement and Objectives**

Understanding how surface water contamination relates to drinking water quality is an important problem in environmental data analysis. In many cases, surface water bodies serve as primary sources for municipal water systems, and changes in these sources may influence the quality of treated drinking water. Identifying patterns in contaminant levels across these systems can provide useful insights for monitoring and management.

The main problem addressed in this project is to analyze contaminant levels in drinking water systems across Kansas and examine whether similar patterns can be observed in surface water data. The goal is to understand which contaminants are most important, how their concentrations change over time, and whether surface water data provides additional context for these patterns.

To address this problem, the project was designed with the following objectives:

- Identify the most frequently reported contaminants in the drinking water dataset and analyze their concentration levels.
- Examine trends in contaminant concentrations over time to understand whether values are increasing, decreasing, or remaining stable.
- Evaluate exceedance patterns by comparing observed values with regulatory limits (MCL) to identify potential risk areas.
- Analyze variability in contaminant data by identifying outliers and understanding extreme values.

- Perform spatial analysis of surface water data to understand how measurements vary across different monitoring locations.
- Focus on nitrate as a key contaminant and analyze its behavior in both drinking water and surface water datasets.
- Explore the possibility of linking surface water contamination with drinking water systems using available data.

During the analysis, it was found that the drinking water dataset does not include geographic identifiers such as latitude, longitude, or a shared location key. This limitation made it impossible to directly connect the two datasets.

Because of this, the project approach was adjusted. Instead of building a direct relationship between the datasets, the analysis was divided into two parts: one focusing on drinking water contaminant patterns and the other focusing on spatial and temporal patterns in surface water.

### **Data Sources and Description**

This project uses two main datasets to analyze water quality patterns in Kansas: a drinking water dataset and a surface water dataset. These datasets were selected to support both statistical analysis and spatial analysis of contaminants.

**Drinking Water Dataset:** The drinking water dataset contains records from municipal water systems, including measurements of different contaminants. Each record represents a reported value for a specific contaminant, along with related information such as unit, regulatory limit, and collection date.

The key variables used in this dataset include:

- Contaminant name
- Reported concentration value (Highest Value)
- Measurement unit (ppm, mg/L, µg/L)
- Maximum Contaminant Level (MCL)
- Collection date and year
- Source or system identifier

One of the main challenges in this dataset was the inconsistency in units and value formats. Concentrations were reported using different units, which made direct comparison difficult. In addition, some values were recorded in non-numeric formats such as “<0.5” and “ND” (non-detect), which required cleaning before analysis.

After preprocessing, all concentration values were converted into a common unit (ppb), and only valid numeric records were retained. Duplicate entries were removed, and the dataset was prepared for further analysis.

**Surface Water Dataset:** The surface water dataset was obtained from the Water Quality Portal (WQP), which provides water quality measurements collected from monitoring locations across the United States.

**The key variables used in this dataset include:**

- Monitoring Location Identifier
- Monitoring Location Name
- Date of measurement
- Characteristic name (e.g., nitrate, pH, turbidity)
- Measurement value
- Unit
- Latitude and longitude

Unlike the drinking water dataset, this dataset includes geographic coordinates, which allows for spatial analysis. This makes it possible to visualize monitoring locations and analyze how contaminant levels vary across different areas.

After cleaning, the surface water dataset contained approximately 17,000 records across 164 monitoring locations. Records with missing geographic or measurement values were removed to ensure reliable analysis.

**Focus on Nitrate:** During initial analysis, nitrate was identified as one of the most important contaminants due to its significantly higher concentration levels compared to others. Because of this, nitrate was selected for detailed analysis across both datasets.

In the surface water dataset, nitrate records were filtered to create a subset of approximately 420 observations across 74 monitoring locations. This subset was used for spatial and temporal analysis.

**Data Limitations:** A key limitation identified in this project is the lack of geographic information in the drinking water dataset. Specifically, the dataset does not include latitude, longitude, or a shared identifier that can be used to link it with surface water monitoring locations.

Because of this, it is not possible to directly connect drinking water systems with specific surface water sources. This limitation affects the ability to perform combined spatial analysis across the two datasets.

**Summary of Data Sources:** In summary, the drinking water dataset provides detailed information on contaminant levels and trends within municipal systems, while the surface water dataset provides geographic context and supports spatial analysis. Although a direct linkage between the datasets is not possible, both datasets contribute to understanding water quality patterns in Kansas.

During this stage, we realized that combining both datasets directly would be difficult due to missing geographic information.

## **Data Preparation**

The data preparation phase was essential for ensuring that both datasets were clean, consistent, and suitable for analysis. Since the datasets were collected from different sources, they contained inconsistencies in units, formats, and missing values that needed to be addressed before meaningful analysis could be performed.

**Drinking Water Data Preparation:** The drinking water dataset required significant preprocessing due to variations in how contaminant values were recorded. One of the main issues was the use of different measurement units such as ppm, mg/L, and µg/L. To make the data comparable across all contaminants, all values were converted into a common unit, parts per billion (ppb). This step ensured consistency and allowed accurate comparison of concentration levels.

Another challenge was handling non-numeric entries such as “<0.5” and “ND” (non-detect). These values were processed by extracting the numeric component where possible or treating them as missing values. This helped prevent errors in calculations and ensured that only valid numeric values were used in analysis.

Duplicate records were removed to avoid repeated observations. In addition, rows with missing or invalid values in key fields such as contaminant name, concentration, and year were excluded. This improved the overall quality and reliability of the dataset.

To support analysis, new variables were created. The standardized concentration value (Highest\_ppb) was used as the primary variable for analysis. Regulatory limits were also converted into ppb (MCL\_ppb) to allow direct comparison with observed values. An exceedance indicator (Exceed\_MCL) was created to identify cases where contaminant levels exceeded regulatory limits.

An outlier detection step was also performed using the interquartile range (IQR) method for each contaminant. This helped identify unusually high or low values, which are important for understanding variability in the data.

### **Surface Water Data Preparation:**

The surface water dataset required a different type of preprocessing focused on preparing the data for spatial analysis. Relevant columns such as measurement value, characteristic name, date, latitude, and longitude were selected for analysis.

Measurement values and geographic coordinates were converted into numeric format to ensure consistency. Records with missing latitude or longitude values were removed, as they cannot be used for mapping or spatial analysis.

Date values were converted into a standard format, and a year variable was extracted to support trend analysis over time. The dataset included many different water quality characteristics, but nitrate was selected for detailed analysis because it showed importance in the drinking water dataset. A filtered subset of nitrate observations was created, which was used for spatial visualization and temporal analysis.

**Summary:** Overall, the data preparation phase ensured that both datasets were accurate, consistent, and ready for analysis. Standardizing units, handling missing values, and creating derived variables allowed for reliable comparisons and meaningful interpretation of results in later stages of the project.

## Methodology

The project was carried out in multiple steps, starting with data cleaning and then moving to analysis.

During the cleaning phase, we faced issues with inconsistent units such as ppm, mg/L, and ppb. To make the data comparable, we converted all values into ppb. We also had to handle values like "<0.5" and "ND", which required careful processing.

After cleaning, we performed exploratory analysis to understand which contaminants appear frequently and how their values are distributed. Based on this, we selected key contaminants for further analysis.

We then tried to analyze trends over time and applied a regression model to check whether year could explain changes in concentration. However, we observed that combining all contaminants in a single model did not produce meaningful results.

For the surface water dataset, we used geographic information to visualize monitoring locations and analyze spatial patterns. We focused on nitrate for deeper analysis since it showed higher values in the drinking water dataset.

Overall, the methodology was adjusted based on the data limitations we encountered during the project.

### **Drinking Water Analysis Approach:**

We started with the drinking water dataset and focused on cleaning and understanding the data. During the cleaning process, we noticed that the dataset had multiple units such as ppm,

mg/L, and ppb. To make the data consistent, we converted all values into ppb. We also handled values like "<0.5" and "ND", which required some manual processing.

After cleaning, we explored the data to identify the most frequently occurring contaminants. Based on this, we selected a few key contaminants for further analysis. We then analyzed trends over time and applied a regression model to check whether year could explain changes in concentration. However, we found that combining all contaminants in one model did not give meaningful results.

### **Surface Water Analysis Approach:**

For the surface water dataset, we used the available geographic information to understand spatial patterns. We cleaned the dataset by converting values into numeric format and removing records with missing latitude and longitude. This allowed us to plot monitoring locations on a map.

We focused on nitrate for detailed analysis since it showed higher values in the drinking water dataset. We analyzed how nitrate values vary across locations and over time. From this, we observed that nitrate values fluctuate and are spread across multiple locations rather than being concentrated in one specific area.

### **Combined Approach:**

Although both datasets were analyzed separately, the methodology was designed to maintain a consistent focus on key contaminants, particularly nitrate.

Overall, the methodology combines data cleaning, exploratory analysis, visualization, and simple modeling techniques to extract meaningful insights while ensuring that the analysis remains aligned with the available data.

## **Drinking Water Analysis**

The drinking water analysis focuses on identifying key contaminants, examining their concentration levels, and understanding how these values change over time. This section presents the main findings based on exploratory data analysis and supporting visualizations.

**Contaminant Frequency Analysis:** The first step in the analysis was to identify the most frequently reported contaminants in the dataset. Understanding frequency is important because it helps determine which contaminants have sufficient data for meaningful analysis.

The results show that Barium, Fluoride, Selenium, Nitrate, and Chromium are the most reported contaminants. These contaminants appear significantly more often than others, making them suitable for further trend and concentration analysis.

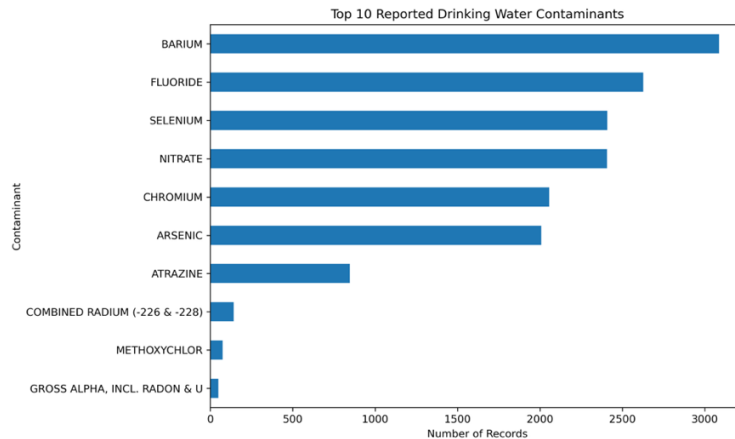


Figure 1: Top 10 Reported Drinking Water Contaminants

**Trend Analysis of Top Contaminants:** After identifying the most frequent contaminants, a trend analysis was performed to examine how their concentration levels change over time. Average concentration values were calculated for each year and plotted for comparison.

The results show that most contaminants, including Barium, Fluoride, Selenium, and Chromium, remain relatively stable over time with only minor variations. In contrast, nitrate shows a significantly different pattern, with much higher concentration levels compared to other contaminants.

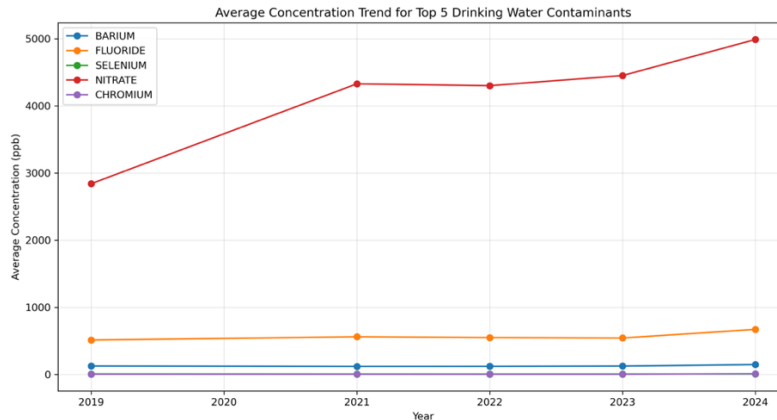


Figure 2: Average Concentration Trends for Top 5 Contaminants

**Nitrate Analysis:** Nitrate was analyzed separately because it showed the highest concentration levels among all contaminants. The trend analysis indicates that nitrate values increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024, with some fluctuations between years.

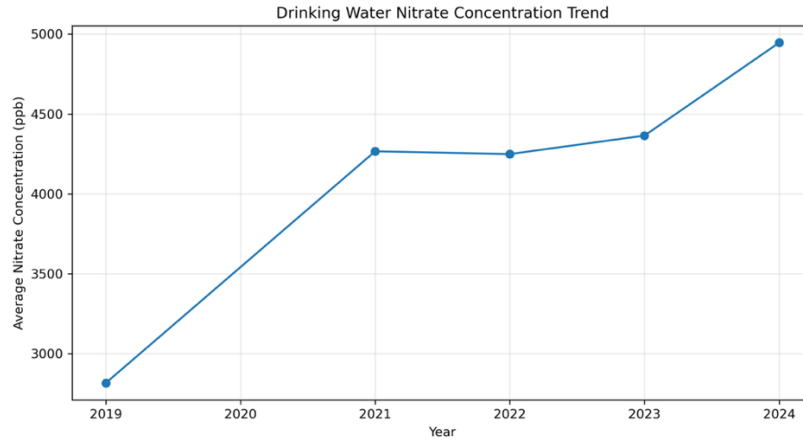


Figure 3: Drinking Water Nitrate Concentration Trend (2019–2024)

This pattern suggests that nitrate behaves differently from other contaminants and may require closer monitoring. The higher concentration levels also make nitrate a key contaminant of interest in this study.

### Regression Analysis

A simple linear regression model was applied to evaluate whether year can explain variation in contaminant concentration when all contaminants are combined. The results show a very low R-squared value and a high p-value, indicating that there is no significant linear relationship.

This outcome suggests that combining all contaminants into a single model is not meaningful, as each contaminant has different behavior and concentration ranges. Therefore, contaminant-specific analysis provides more reliable insights.

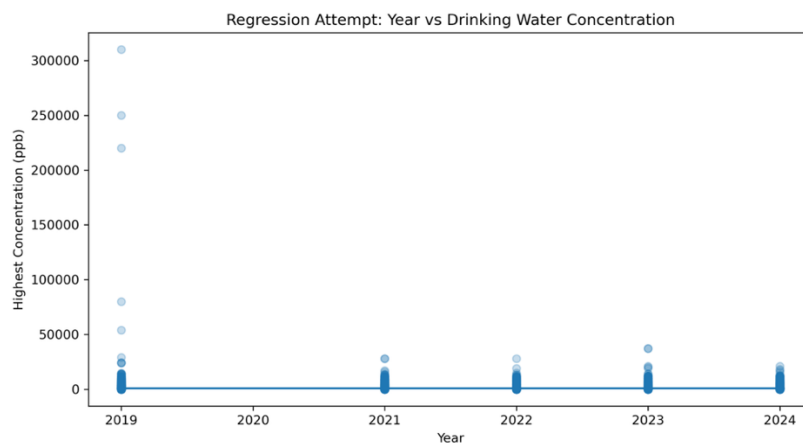


Figure 4: Regression Analysis (Year vs Concentration)

## Exceedance Analysis

Exceedance analysis was conducted by comparing observed contaminant values with regulatory limits (MCL). This helps identify contaminants that may pose potential risks.

The results show that some contaminants have high exceedance rates but are based on a small number of observations. In contrast, nitrate has a moderate exceedance rate but a large number of records, making it more significant in terms of overall impact.

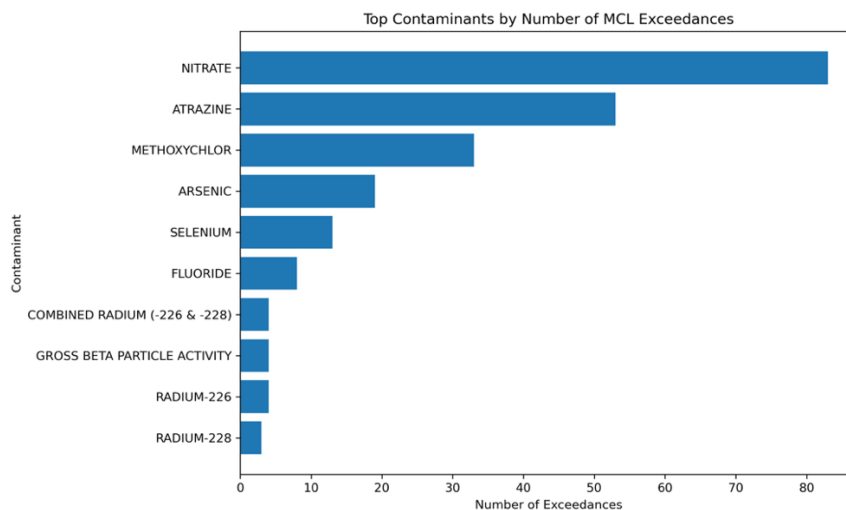


Figure 5: Contaminants by Number of MCL Exceedances

## Summary of Drinking Water Analysis

Overall, the drinking water analysis shows that while several contaminants are commonly reported, nitrate stands out due to its higher concentration levels and noticeable variation over time. Most other contaminants remain relatively stable, indicating less variability.

The regression analysis confirms that aggregated modeling is not effective, and individual contaminant analysis is more meaningful. These findings highlight nitrate as a key contaminant that requires attention in further analysis.

Key findings from drinking water analysis:

- Nitrate shows significantly higher concentration levels compared to other contaminants.
- Most contaminants remain relatively stable over time.
- Aggregated regression does not provide meaningful insights.

- Exceedance patterns vary depending on both rate and number of observations.

## Surface Water Analysis

The surface water analysis focuses on understanding how water quality measurements vary across different locations and over time. Unlike the drinking water dataset, this dataset includes geographic coordinates, which allows for spatial analysis of monitoring locations.

### Monitoring Location Distribution

The first step in the surface water analysis was to visualize the distribution of monitoring locations. This helps understand how data is geographically spread across the state and whether certain areas have more coverage than others.

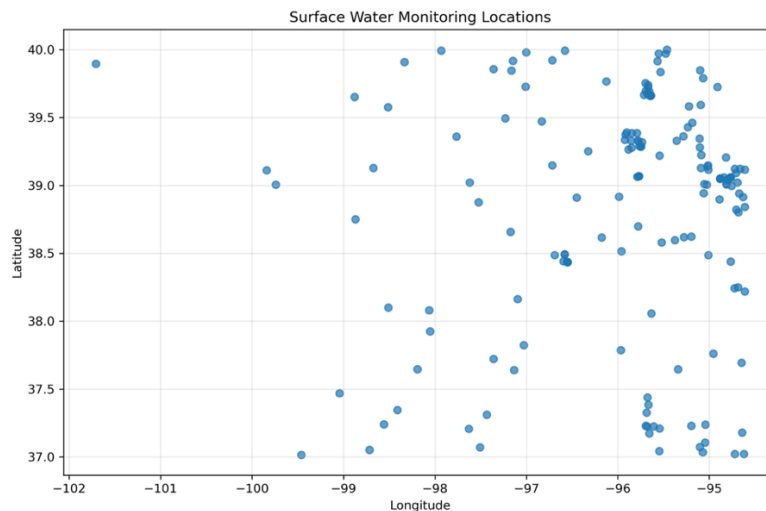


Figure 6: Distribution of Surface Water Monitoring Locations

The plot shows that monitoring locations are distributed across Kansas, although the distribution is not uniform. Some regions have a higher concentration of monitoring sites, while others have fewer. This variation in coverage should be considered when interpreting spatial results.

### Surface Water Characteristics:

The dataset includes a wide range of measured characteristics such as turbidity, pH, dissolved oxygen, temperature, and nitrate. These variables provide a general view of water quality conditions across monitoring sites.

Understanding the most frequently measured characteristics helps identify which variables are suitable for detailed analysis. Among these, nitrate was selected for further investigation because of its importance in the drinking water analysis.

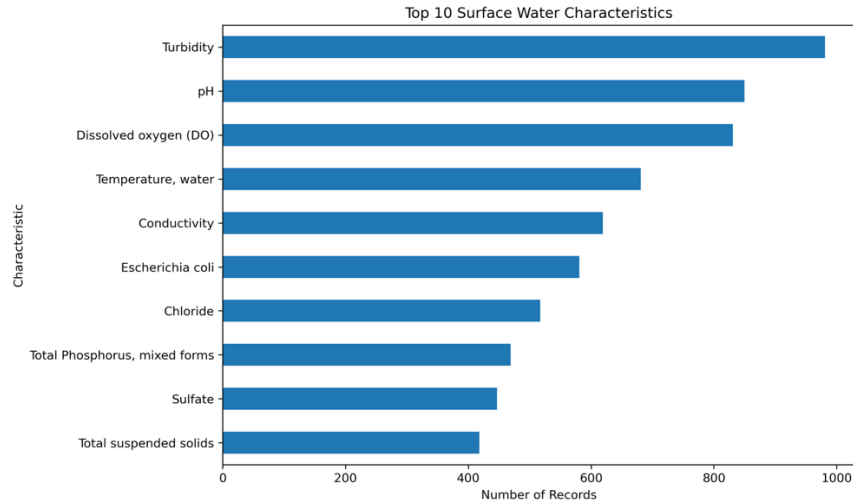


Figure 7: Most Frequently Measured Surface Water Characteristics

### Nitrate Spatial Analysis

A subset of nitrate observations was created to analyze its spatial distribution. The dataset contains approximately 420 nitrate measurements across 74 monitoring locations. The spatial plot shows that nitrate values vary across different locations. Some sites have higher concentration values, while others remain lower. However, there is no single region where high nitrate values are consistently concentrated. This indicates that nitrate contamination is distributed across multiple locations rather than being limited to a specific hotspot.

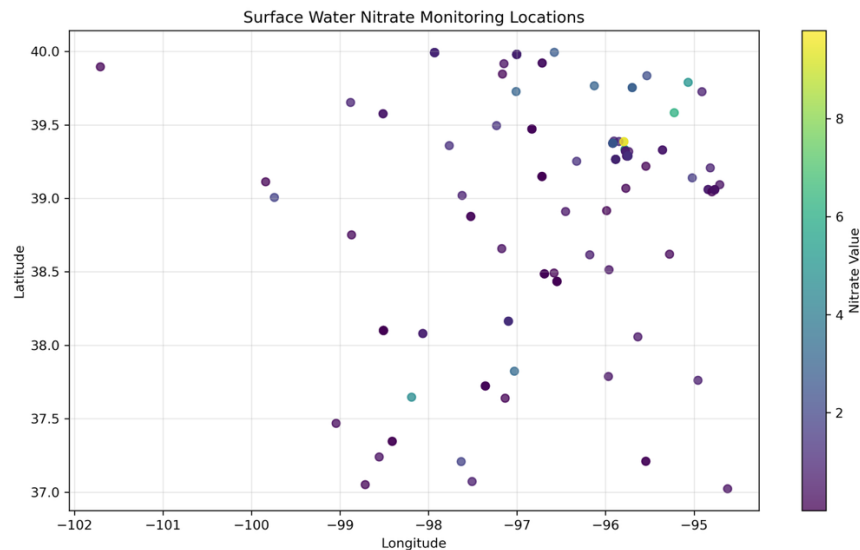


Figure 8: Spatial Distribution of Nitrate in Surface Water

### Nitrate Temporal Analysis

To understand how nitrate levels change over time, a yearly trend analysis was performed using average values. The results show that nitrate concentrations fluctuate between approximately 0.8 and 1.7 across different years.

There is no clear increasing or decreasing trend, and the variation suggests that nitrate levels in surface water are influenced by changing environmental conditions rather than a consistent pattern over time.

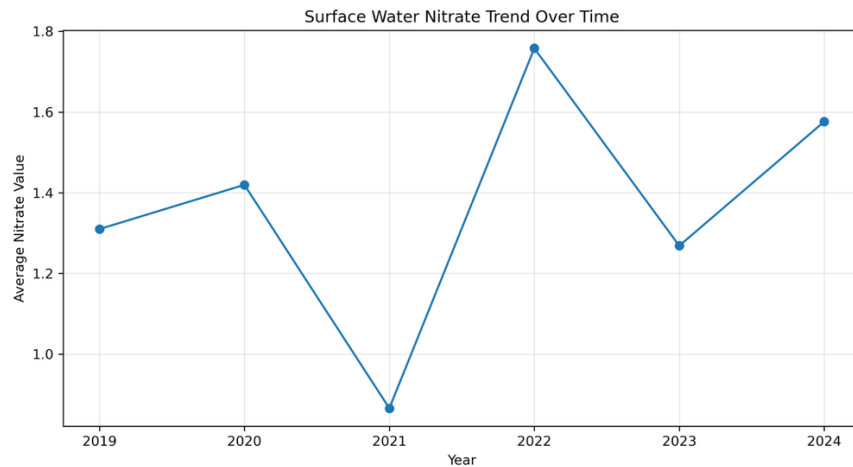


Figure 9: Surface Water Nitrate Trend Over Time

### Site-Level Nitrate Analysis

A site-level summary was created to examine how nitrate values differ across monitoring locations. This analysis helps identify whether certain sites consistently show higher concentration levels.

The results indicate that some locations have higher average nitrate values than others. However, these higher values are not concentrated in a specific region, which supports the earlier observation that nitrate is distributed across multiple locations.

| MonitoringLocationID      | Records | Mean_Value  | Median_Value | Max_Value | Latitude    | Longitude    |
|---------------------------|---------|-------------|--------------|-----------|-------------|--------------|
| PRAIRIEBAND-MW11          | 2       | 8.8         | 8.8          | 9.8       | 39.3858     | -95.7932     |
| PRAIRIEBAND-MW08          | 2       | 6.7         | 6.7          | 8.6       | 39.3318     | -95.7814     |
| NARS_WQX-NRS_KS-10435     | 1       | 6.19        | 6.19         | 6.19      | 39.58314228 | -95.22446835 |
| NARS_WQX-NRS_KS-10449     | 1       | 5.02        | 5.02         | 5.02      | 39.78982916 | -95.06722268 |
| NARS_WQX-NRS_KS-10249     | 1       | 4.65        | 4.65         | 4.65      | 37.64636548 | -98.19185293 |
| NARS_WQX-NRS_KS-10248     | 1       | 3.3074      | 3.3074       | 3.3074    | 37.82311746 | -97.0303315  |
| NARS_WQX-NRS_KS-10210     | 2       | 3.1957      | 3.1957       | 3.6902    | 39.75447276 | -95.69903262 |
| NARS_WQX-NRS_KS-10368     | 1       | 3.1         | 3.1          | 3.1       | 39.72738834 | -97.01089354 |
| NARS_WQX-NRS_KS-10404     | 1       | 3.01        | 3.01         | 3.01      | 39.76648562 | -96.12754016 |
| NARS_WQX-NRS_KS-10205     | 1       | 2.7479      | 2.7479       | 2.7479    | 39.99441929 | -96.57905059 |
| NARS_WQX-NRS_KS-10443     | 1       | 2.26        | 2.26         | 2.26      | 39.83518293 | -95.53318623 |
| NARS_WQX-NRS_KS-10468     | 1       | 1.84        | 1.84         | 1.84      | 39.00631846 | -99.74227905 |
| NARS_WQX-NRS_KS-10367     | 1       | 1.84        | 1.84         | 1.84      | 37.20753864 | -97.63141656 |
| 21NEB001_WQX-SLB1BLUE000  | 155     | 1.639869677 | 1.73         | 7.55      | 39.980149   | -97.003893   |
| NARS_WQX-NWC_KS-10130     | 1       | 1.63        | 1.63         | 1.63      | 39.139578   | -95.020855   |
| 21NEB001_WQX-SRE1REPUB000 | 87      | 1.486218391 | 1.25         | 3.74      | 39.992569   | -97.931967   |
| NARS_WQX-NRS_KS-10050     | 1       | 1.3473      | 1.3473       | 1.3473    | 39.49522569 | -97.23190219 |
| NARS_WQX-NRS_KS-10046     | 1       | 1.3143      | 1.3143       | 1.3143    | 39.25252179 | -96.32659742 |
| NARS_WQX-NRS_KS-10360     | 1       | 1.27        | 1.27         | 1.27      | 39.35926743 | -97.76596877 |
| PRAIRIEBAND-SCO7          | 18      | 1.151111111 | 1.05         | 3.3       | 39.2658     | -95.8864     |
| PRAIRIEBAND-SCO3          | 25      | 1.1228      | 0.83         | 2.9       | 39.3758     | -95.9183     |
| PRAIRIEBAND-LSCO6         | 10      | 1.107       | 1.15         | 1.8       | 39.2888     | -95.7483     |
| PRAIRIEBAND-MW10          | 1       | 1.1         | 1.1          | 1.1       | 39.3873     | -95.8494     |
| NARS_WQX-NRS_KS-10269     | 2       | 1.03485     | 1.03485      | 2.0217    | 39.92164243 | -96.71631982 |
| PRAIRIEBAND-MW09          | 1       | 0.97        | 0.97         | 0.97      | 39.3904     | -95.9053     |
| NARS_WQX-NRS_KS-10255     | 1       | 0.9048      | 0.9048       | 0.9048    | 39.65317317 | -98.88144118 |
| NARS_WQX-NRS_KS-10226     | 2       | 0.83935     | 0.83935      | 1.1567    | 39.32949678 | -95.35688326 |
| NARS_WQX-NRS_KS-10208     | 1       | 0.8141      | 0.8141       | 0.8141    | 39.01991425 | -97.62109954 |
| NARS_WQX-NRS_KS-10222     | 1       | 0.8079      | 0.8079       | 0.8079    | 39.20750693 | -94.81716042 |
| NARS_WQX-NRS_KS-10039     | 1       | 0.7685      | 0.7685       | 0.7685    | 39.72624414 | -94.91122091 |
| NARS_WQX-NRS_KS-10431     | 1       | 0.724       | 0.724        | 0.724     | 38.6157999  | -96.17854363 |
| NARS_WQX-NRS_KS-10246     | 1       | 0.7193      | 0.7193       | 0.7193    | 37.76116475 | -94.9562043  |
| NARS_WQX-NRS_KS-10199     | 1       | 0.6977      | 0.6977       | 0.6977    | 37.07185357 | -97.50849573 |
| EPA_R7_WQX-RSTIP09        | 3       | 0.695666667 | 0.27         | 1.55      | 38.163689   | -97.097927   |
| NARS_WQX-NRS_KS-10051     | 2       | 0.6841      | 0.6841       | 0.7043    | 39.06027253 | -94.84194972 |
| PRAIRIEBAND-LSCO3         | 8       | 0.6675      | 0.665        | 1.2       | 39.3225     | -95.7758     |
| NARS_WQX-NRS_KS-10196     | 2       | 0.6445      | 0.6445       | 0.74      | 39.57629487 | -98.51414655 |
| PRAIRIEBAND-MW03          | 1       | 0.61        | 0.61         | 0.61      | 39.318614   | -95.738642   |

| MonitoringLocationID  | Records | Mean_Value | Median_Value | Max_Value | Latitude    | Longitude    |
|-----------------------|---------|------------|--------------|-----------|-------------|--------------|
| PRAIRIEBAND-LSCO4     | 4       | 0.57       | 0.6          | 0.63      | 39.2887     | -95.7617     |
| NARS_WQX-NRS_KS-10264 | 1       | 0.5171     | 0.5171       | 0.5171    | 38.91041431 | -96.45214179 |
| NARS_WQX-NWC_KS-10102 | 2       | 0.5151     | 0.5151       | 0.991     | 38.07983    | -98.06508    |
| NARS_WQX-NRS_KS-10250 | 1       | 0.5087     | 0.5087       | 0.5087    | 38.05708438 | -95.63541799 |
| EPA_R7_WQX-KVC5       | 1       | 0.491      | 0.491        | 0.491     | 39.05908628 | -94.76684306 |
| NARS_WQX-NWC_KS-10143 | 1       | 0.478      | 0.478        | 0.478     | 37.787233   | -95.967877   |
| NARS_WQX-NRS_KS-10293 | 1       | 0.4029     | 0.4029       | 0.4029    | 38.51416207 | -95.96179912 |
| EPA_R7_WQX-CON2       | 1       | 0.374      | 0.374        | 0.374     | 39.093076   | -94.71003265 |
| NARS_WQX-NRS_KS-10289 | 1       | 0.3634     | 0.3634       | 0.3634    | 37.02237885 | -94.6211974  |
| EPA_R7_WQX-RSTIP10    | 4       | 0.3575     | 0.35         | 0.71      | 37.722646   | -97.358462   |
| NARS_WQX-NRS_KS-10252 | 1       | 0.332      | 0.332        | 0.332     | 38.65767114 | -97.17331128 |
| NARS_WQX-NWC_KS-10129 | 1       | 0.331      | 0.331        | 0.331     | 37.239256   | -98.557535   |
| NARS_WQX-NRS_KS-10361 | 1       | 0.33       | 0.33         | 0.33      | 39.91711927 | -97.14841713 |
| EPA_R7_WQX-RSTIP03    | 5       | 0.25282    | 0.132        | 0.768     | 39.472247   | -96.832223   |
| NARS_WQX-NRS_KS-10448 | 1       | 0.215      | 0.215        | 0.215     | 37.4680573  | -99.04453627 |
| EPA_R7_WQX-KVC4       | 1       | 0.196      | 0.196        | 0.196     | 39.05770136 | -94.76781834 |
| NARS_WQX-NRS_KS-10245 | 1       | 0.189      | 0.189        | 0.189     | 38.75125952 | -98.87007228 |
| NARS_WQX-NRS_KS-10262 | 1       | 0.1881     | 0.1881       | 0.1881    | 38.61995291 | -95.27727329 |
| NARS_WQX-NRS_KS-10093 | 2       | 0.1469     | 0.1469       | 0.1822    | 37.20970471 | -95.54539085 |
| EPA_R7_WQX-RSTIP04    | 2       | 0.1435     | 0.1435       | 0.153     | 38.877043   | -97.52317    |
| EPA_R7_WQX-KVC6       | 1       | 0.139      | 0.139        | 0.139     | 39.060303   | -94.7665188  |
| EPA_R7_WQX-RSTIP08    | 5       | 0.0956     | 0.131        | 0.154     | 38.486057   | -96.691025   |
| NARS_WQX-NRS_KS-10446 | 1       | 0.079      | 0.079        | 0.079     | 39.84655733 | -97.16511325 |
| EPA_R7_WQX-KVC2       | 1       | 0.0752     | 0.0752       | 0.0752    | 39.0444992  | -94.79866752 |
| EPA_R7_WQX-RSTIP05    | 10      | 0.065      | 0.02         | 0.147     | 38.10035    | -98.509398   |
| EPA_R7_WQX-RSTIP02    | 5       | 0.0606     | 0.02         | 0.223     | 39.149088   | -96.718446   |
| EPA_R7_WQX-RSTIP07    | 10      | 0.0591     | 0.02         | 0.219     | 38.43461    | -96.54928    |
| NARS_WQX-NRS_KS-10290 | 1       | 0.0548     | 0.0548       | 0.0548    | 37.05079383 | -98.71547714 |
| NARS_WQX-NRS_KS-10235 | 1       | 0.051      | 0.051        | 0.051     | 39.89576845 | -101.7077863 |
| EPA_R7_WQX-LBB2       | 1       | 0.0442     | 0.0442       | 0.0442    | 39.0685004  | -95.77104932 |
| NARS_WQX-NRS_KS-10412 | 1       | 0.03       | 0.03         | 0.03      | 38.91634574 | -95.9883252  |
| NARS_WQX-NRS_KS-10345 | 1       | 0.01       | 0.01         | 0.01      | 39.21883246 | -95.54474928 |
| NARS_WQX-NRS_KS-10484 | 1       | 0.008      | 0.008        | 0.008     | 38.49167    | -96.58222    |
| NARS_WQX-NWC_KS-10098 | 2       | 0.0072     | 0.0072       | 0.0135    | 37.34562    | -98.41065    |
| NARS_WQX-NRS_KS-10271 | 1       | 0.0054     | 0.0054       | 0.0054    | 39.11269578 | -99.84059319 |
| NARS_WQX-NWC_KS-10142 | 1       | 0.0016     | 0.0016       | 0.0016    | 37.639357   | -97.134782   |

Table 1: Summary of Nitrate Values by Monitoring Location

Overall, the surface water analysis shows that monitoring locations are spread across Kansas and provide sufficient coverage for spatial analysis. Nitrate values vary across locations and fluctuate over time, with no clear linear trend.

The spatial and site-level analysis indicates that nitrate contamination is not limited to a single hotspot but is distributed across multiple monitoring locations. These findings provide useful context for understanding contaminant behavior in surface water systems.

Key findings from surface water analysis:

- Monitoring locations are distributed across Kansas with uneven coverage.
- Nitrate concentrations vary across locations with no single hotspot.
- Temporal trends show fluctuations rather than a consistent pattern.
- Site-level analysis confirms variation across multiple locations.

## Comparative Discussion

This section compares the key findings from the drinking water and surface water analyses to understand how contaminant patterns differ and where they show similarities. Although a direct linkage between the datasets is not possible, comparing the results provides useful insights into overall water quality behavior.

From the drinking water analysis, nitrate was identified as the most significant contaminant based on its higher concentration levels and noticeable variation over time. The trend showed an increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. In contrast, most other contaminants such as Barium, Fluoride, Selenium, and Chromium remained relatively stable over time.

In the surface water analysis, nitrate was also examined as the primary contaminant of interest. However, the behavior observed in surface water was different. Instead of showing a clear increasing trend, nitrate values fluctuated between approximately 0.8 and 1.7 across different years. This indicates that surface water nitrate levels vary over time but do not follow a consistent pattern.

Spatial analysis further highlighted differences in how nitrate appears in surface water compared to drinking water. In surface water, nitrate concentrations were distributed across multiple monitoring locations, with no single region consistently showing the highest values. This suggests that contamination is spread across different areas rather than being concentrated in a specific hotspot.

While both datasets highlight nitrate as an important contaminant, the patterns observed in each system are not directly comparable due to differences in data structure, measurement context, and lack of shared identifiers. The drinking water dataset reflects treated water supplied to communities, while the surface water dataset represents raw environmental measurements.

A key limitation in this comparison is the absence of geographic information in the drinking water dataset. Without latitude, longitude, or a shared location identifier, it is not possible to match specific drinking water systems with nearby surface water monitoring sites. Because of this, the analysis cannot establish a direct relationship between contamination in surface water and drinking water quality.

Despite this limitation, comparing the two analyses provides useful context. The presence of nitrate in both datasets suggests that it is a widespread contaminant of concern. However, the differences in trends and spatial patterns indicate that additional data would be required to fully understand how surface water contamination influences drinking water systems.

Overall, this comparison highlights the importance of data availability and structure when attempting to analyze interconnected environmental systems. It also demonstrates that careful interpretation is necessary to avoid drawing unsupported conclusions.

## Limitations

During this project, we faced a few limitations that affected how the analysis was performed.

- The drinking water dataset does not include geographic information such as latitude or longitude. Because of this, we were not able to directly link drinking water systems with surface water monitoring locations.
- The two datasets represent different contexts. Drinking water data reflects treated water supplied to communities, while surface water data represents raw environmental measurements, making direct comparison difficult.
- A significant amount of time was spent handling inconsistent formats and units such as ppm, mg/L, and ppb, along with values like "<0.5" and "ND".
- Some records contained missing or incomplete values, which required cleaning and removal of certain observations.
- The analysis focused mainly on a few key contaminants, especially nitrate, rather than covering all contaminants in the dataset.
- External factors such as rainfall, seasonal changes, and land use were not included, which may influence contaminant levels.

Despite these limitations, the analysis still provides useful insights based on the available data.

## Future Work and Recommendations

Based on the findings of this project, the following improvements can be considered for future work:

- Include geographic identifiers for drinking water systems to enable direct linkage with surface water data.
- Incorporate additional environmental variables such as rainfall, land use, and seasonal factors to better explain variations in contaminant levels.
- Expand the analysis to include a broader set of contaminants beyond nitrate.
- Apply more advanced modeling techniques if more detailed and connected data becomes available.
- Improve data consistency and standardization across sources to support integrated analysis.

These improvements can help provide a better understanding of how surface water and drinking water systems are connected.

## Conclusion

In this project, we analyzed drinking water and surface water datasets to understand contaminant patterns in Kansas. From the analysis, we observed that nitrate has higher concentration levels compared to other contaminants and shows noticeable variation over time.

We also found that most other contaminants remain relatively stable. When we applied regression, the results showed that year alone does not explain the variation in contaminant values when all contaminants are combined.

In the surface water analysis, nitrate values varied across locations and changed over time, but there was no clear trend. We also noticed that higher values are spread across different locations instead of being concentrated in a single area.

One important limitation of this project was the lack of geographic information in the drinking water dataset, which made it impossible to directly connect it with surface water data.

Overall, this project gave us a better understanding of how water quality data can be analyzed and how real-world limitations affect the results.

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